

The corresponding p-values are reported at the bottom of table 4. In the OLS regression, the test rejects the hypothesis that the coefficient on the test variable is zero. This means that the model estimated using the logarithmic specification is inappropriate. A similar result is found for the OLS estimated using $\ln(1 + T_{ij})$ as the dependent variable and NLS. In contrast, the models estimated using the Poisson regressions pass the RESET test, i.e., the RESET test provides no evidence of misspecification of the gravity equations estimated using the PPML. With this particular specification, the model estimated using Tobit also passes the test for the traditional gravity equation.

Finally, we also check whether the particular pattern of heteroskedasticity assumed by the models is appropriate. As explained in section 3.2, the adequacy of the log-linear model was checked using the Park-type test, whereas the hypothesis $V[y_i|x] \propto \mu(x_i\beta)$ was tested by evaluating the significance of the coefficient on the term $\ln(\tilde{y}_i) \sqrt{\tilde{y}_i}$ in the Gauss-Newton regression indicated in expression (13). The p-values of the tests are reported in Table 4. Again, the log-linear specification is unequivocally rejected. On the other hand, these results indicate that the estimated coefficient on $\ln(\tilde{y}_i) \sqrt{\tilde{y}_i}$ is insignificantly different from zero at the usual 5 percent level. This implies that the Poisson PML assumption, $V[y_i|x] = \lambda_0 E[y_i|x]$ cannot be rejected at this significance level.

Table 4: Results of the tests for type of heteroskedasticity (p-values).

Test (Null hypothesis)	Trade > 0	Full sample
GNR ($V[y_i x] \propto \mu(x_i\beta)$)	0.144	0.115
Park (OLS is valid)	0.000	0.000

5.2.2 The Anderson-van Wincoop Gravity Equation

Table 5 presents the estimated coefficients for the Anderson-van Wincoop (2003) gravity equation, which controls more properly for multilateral resistance terms by introducing exporter and importer specific effects.

As before, the columns show, respectively, the estimated coefficients obtained using OLS on the log of trade, OLS on $\ln(1 + T_{ij})$, Tobit, NLS, PPML on the positive-trade

To sum up, whether or not fixed effects are used in the specification of the model, we find strong evidence that estimation methods based on the log-linearization of the gravity equation suffer from severe misspecification, which hinders the interpretation of the results. NLS is also generally unreliable. In contrast, the models estimated by PPML show no signs of misspecification and, in general, do not produce the puzzling results generated by the other methods.²⁹

6. Conclusions

In this paper, we argue that the standard empirical methods used to estimate gravity equations are inappropriate. The basic problem is that log-linearization (or, indeed, any non-linear transformation) of the empirical model in the presence of heteroskedasticity leads to inconsistent estimates. This is because the expected value of the logarithm of a random variable depends on higher-order moments of its distribution. Therefore, if the errors are heteroskedastic, the transformed errors will be generally correlated with the covariates. An additional problem of log-linearization is that it is incompatible with the existence of zeroes in trade data, which led to several unsatisfactory solutions, including truncation of the sample (that is, elimination of zero-trade pairs) and further non-linear transformations of the dependent variable.

²⁹It is worth noting that the large differences in estimates among the various methods persist when we look at a smaller subsample of countries that account for most of world trade and, quite likely, have better data. More specifically, we run similar regressions for the subsample of 63 countries included in Frankel (1997)'s study. These countries accounted for almost 90 percent of the world trade reported to the United Nations in 1992. One advantage of this subsample is that the number of zeroes is significantly reduced. Heteroskedasticity, however, is still a problem: The null hypothesis of homoskedasticity is rejected in both the traditional and the fixed-effects gravity equations. As before, PPML generates a smaller role for distance and common language than OLS, and, unlike OLS, PPML predicts no role for colonial ties. In line with the findings documented in Frankel (1997), the OLS estimated coefficient on the free-trade-agreement dummy is negative in both specifications of the gravity equation, whereas PPML predicts a positive and significant effect (slightly bigger than that found for the whole sample. This results are available – at request – from the authors.